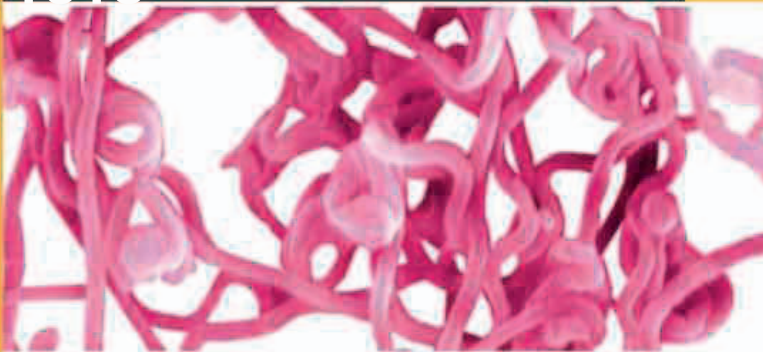


65 THOUSAND

THE NUMBER OF SPECIMENS UNEARTHED BY WALCOTT AT BURGESS SHALE



The distinctive corkscrew shaped bacteria that cause syphilis were finally conquered by an arsenic-based drug initially known as 606.

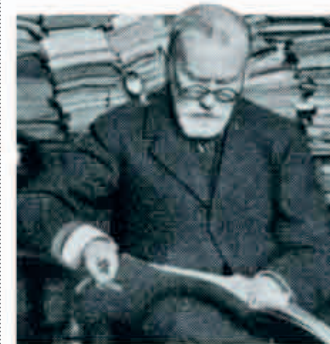
ON MAY 20, HALLEY'S COMET

came closer to Earth than at any time since 1835. *The New York Times* had warned of an imminent apocalypse as astronomers had described the comet's poisonous cyanide-containing tail. However, the event passed without disaster.

The Danish and American astronomers, Ejnar Hertzsprung (1873–1967) and Henry Russell (1877–1957), published the **Hertzsprung–Russell, or H-R, diagram** they devised to **classify star types**. The H-R diagram is a scatter graph that plots the relationship between **temperature, luminosity, and size of stars** and distinguishes types of stars as clusters within the chart—white dwarfs, main sequence, super-giants, and red giants. It remains a standard astronomical tool today.

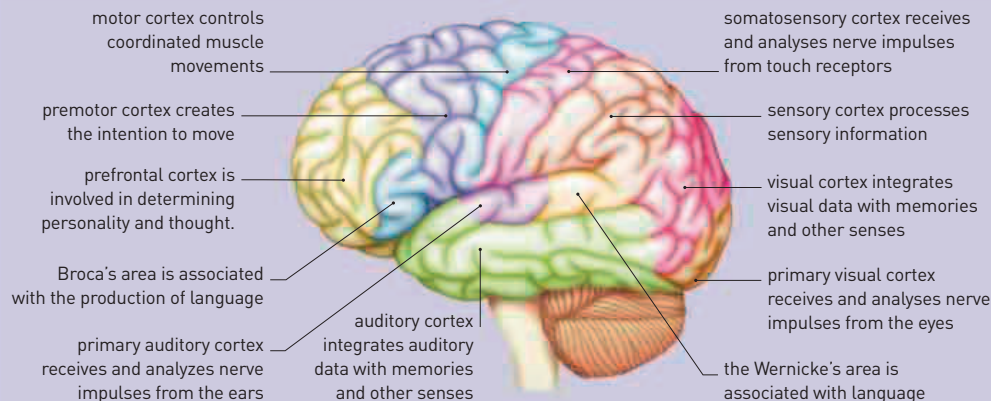
In April, Paul Ehrlich announced the **completion of his arsenic-based drug, 606**, to treat syphilis. By November, the German drug company Hoeschst AG had begun marketing it as **Salvarsan**. Demand for this new drug grew quickly as the treatment was more effective than any previous medication. However, the toxic dangers of its arsenic component remained a concern and, 30 years later, Salvarsan would be replaced by antibiotics (see 1940).

American physicist **Robert W. Wood** (1868–1955)—an expert in optics—was the **first to use infrared (IR) and ultraviolet (UV) radiation to produce photographs**, and published the first examples. Wood pioneered this type of photography and the technology would lead to the modern-day black lights that emit UV radiation and minimal visible light.



PAUL ERLICH (1854–1915)

Starting his career in the medical study of blood, Paul Ehrlich developed stains for revealing cells—including bacteria. From these studies, he pursued the “magic bullet”—a drug that could target a specific infectious organism—initiating the concept of chemotherapy. He also worked on immunization and proposed a theory that explained immune response.



BRAIN FUNCTIONS

This “map” of the most complex part of the brain came out of the painstaking work of Korbinian Brodmann. The surface of the cerebral hemispheres—called the cerebral cortex—is divided into sensory, motor, and association areas.

Sensory areas receive signals from the rest of the body and motor areas control the dispatch of signals to muscles. Association areas are those involved in complicated, higher processing, such as decision-making and language.

Coming to the end of fieldwork at **Burgess Shale** in the Canadian Rocky Mountains, Walcott found a large deposit of fossils. Subsequent investigation revealed it to be one of the

oldest and best-preserved fossil sites, with specimens that dated back 500 million years. Walcott attempted to classify the animals into known groups, but scientists have since discovered that many of them belonged to ancient evolutionary dead ends.

German neuroscientist **Korbinian Brodmann**

(1868–1918) had been working on the fine structure of the part of the brain called the cerebral cortex, which controls higher functions—for example, decision-making and emotion. Using microscopic studies, Brodmann managed to **identify the different functional regions of this part of the brain** that could be linked to processes demonstrated experimentally by other scientists. Brodmann's cerebral map formed the basis for modern understanding of higher brain function.

1,000,000
HYDROGEN IONS IN
HYDROCHLORIC ACID

1 hydrogen ion in water

Comparing pH value

Acidity is determined by hydrogen ions. The concentration of hydrogen ions in hydrochloric acid is million times greater than that of water.

June 10 Paul Ehrlich and Sahachiro Hata file a patent for an arsenic-based antibacterial drug, “606”

June 17 Ernest Rutherford describes how gold foil scatters alpha particles

August French bacteriologist **Charles Nicolle** demonstrates that body lice transmit typhus fever

December 7 US Patent Office registers Leo Baekeland's heat-and-pressure method for making a synthetic plastic, later called **Bakelite**

June 10 Paul Ehrlich and Sahachiro Hata file a patent for an arsenic-based antibacterial drug, “606”

July 25 Louis Blériot flies across the English Channel in the Blériot XI

September **Charles Walcott** discovers the Burgess Shale fossil site in Canada

December Danish botanist **Wilhelm Johannsen** coins the word “gene”

January 13 American inventor **Lee De Forest** broadcasts first live musical performance on radio

June 21 French physician **Jean Hyacinthe Vincent** announces first effective typhoid vaccine

October Robert W. Wood publishes the first **infrared photographs**

July American biologist **Thomas Hunt Morgan** describes inheritance in fruit flies

September 5 Polish physicist **Marie Curie** announces isolation of radium

December Dutchman **Lieutenant van Steyn van Hensbroek** proves existence of the Komodo Dragon